

The Life-like rule B37/S2378: an exhaustive census of 4×4 seeds, disk-shaped growth, and two oscillators with machine-checked periods

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Abstract

The outer-totalistic rule B37/S2378 (a dead cell is born with 3 or 7 live neighbours; a live cell survives with 2, 3, 7 or 8) is a close relative of DryLife (B37/S23). We report the first census of it that we could find. Every one of the 51,472 seeds whose bounding box is exactly 4×4 was evolved until it recurred up to translation; the same protocol was run for Conway's B3/S23 as a control. The B37/S2378 census yields 96 objects, including an oscillator of period 10 (11 cells) and one of period 32 (13 cells), neither of which exists under B3/S23. Seeds that do not settle behave very differently in the two rules: under B37/S2378, 338 of 471 sampled unsettled seeds are still growing after 1,000 generations, as a disk whose interior holds the density of random soup (0.111–0.130) and whose edge advances at 0.056–0.065 c ; under B3/S23 none of the 465 sampled unsettled seeds keeps growing (the 6 still growing at 1,500 generations all settle by 20,000). The periods of both oscillators, the minimality of the period 10, and the failure of the period-10 pattern under Conway's rule are proved in Lean 4 by kernel computation, with no axioms beyond the standard ones.

1 The rule

A Life-like rule is written Bx/Sy : a dead cell with n live Moore neighbours becomes live if $n \in x$, and a live cell stays live if $n \in y$. Conway's Game of Life is B3/S23 [1]. The rule studied here adds birth on 7 and survival on 7 and 8. Its nearest named relative is DryLife, B37/S23, whose explosive behaviour is known in the community [2]; survival on 7 and 8 lets crowded regions persist instead of thinning out. Eppstein [5] discusses which Life-like rules can support both growth and decay.

2 Method

Exhaustive census. Of the $2^{16} - 1 = 65,535$ non-empty patterns of a 4×4 box, the 14,063 that do not touch all four sides are translates of patterns of a smaller box and are left out; each of the remaining 51,472, whose bounding box is exactly 4×4 , is evolved alone in the plane (on a board with a margin larger than the number of generations, so the boundary is never reached) for up to 100 generations. The whole pattern is compared, after cropping, with every earlier generation; the first exact repetition up to translation fixes the outcome: a still life (period 1, no shift), an oscillator (period $p > 1$, no shift) or a spaceship (a shift). A seed that dies is *extinct*; a seed that has not repeated by the last generation is *unsettled*. The settled final pattern is split into objects: live cells at Chebyshev distance at most 3 belong to the same object, so objects close enough to interact are counted together. Objects are identified up to the eight symmetries of the square and, for oscillators and spaceships, across all their phases.

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Soups. 40 random soups of side 128 on a torus, initial density 0.30, evolved for 600 generations; we record the live-cell density.

Growth of unsettled seeds. One unsettled seed in every 25 (B37/S2378) or every 20 (B3/S23) is evolved further, to 1,000 and 1,500 generations respectively, and classified as growing (population still rising in the last window), bounded, or settling. A subsample of the growing seeds is followed further still (5,000 and 20,000 generations), removing escaping gliders once they are more than 20 cells from the core so that they do not inflate the bounding box.

The front. Three growing seeds are evolved for 2,400 generations on a $1,600 \times 1,600$ board. The board is divided into 16×16 blocks; the *core* is the set of blocks with density above 0.03 (a lone glider in a block has density $5/256 < 0.02$), its radius is $\sqrt{\text{area}/\pi}$ and its density is the mean over its blocks.

3 Results

3.1 The census

Table 1 compares the two rules. Of the 51,472 seeds, 15,335 die out and 11,760 (22.8%) are unsettled under B37/S2378, against 15,492 and 9,284 (18.0%) under B3/S23. The only spaceship reached from a 4×4 seed is, in both rules, the glider, which B37/S2378 inherits unchanged. The two columns differ in the oscillators: B37/S2378 has one of period 10 and one of period 32, each reached from 8 seeds, and neither occurs under B3/S23.

class	B37/S2378		B3/S23	
	objects	occurrences	objects	occurrences
still life	66	27,917	55	22,676
oscillator, $p = 2$	26	9,000	24	7,972
oscillator, $p = 3$	1	24	1	48
oscillator, $p = 10$	1	8	–	–
oscillator, $p = 32$	1	8	–	–
spaceship, $p = 4$ (glider)	1	752	1	728
total	96	37,709	81	31,424

Table 1: Objects reached from the 51,472 seeds whose bounding box is exactly 4×4 . *Objects* counts distinct objects up to symmetry and phase; *occurrences* counts how many times each appears in the final patterns (a seed can end in several objects). Cells within Chebyshev distance 3 belong to the same object.

3.2 Two oscillators

Figure 1 shows one phase of each oscillator; their RLE follows.

```
#N p10 oscillator of B37/S2378 (11 cells)
x = 6, y = 5, rule = B37/S2378
bo2bo$o3bo$obo2bo$o3bo$bo2bo!
```

```
#N p32 oscillator of B37/S2378 (13 cells)
x = 6, y = 4, rule = B37/S2378
2b2o$bo2b2o$o3b2o$b5o!
```

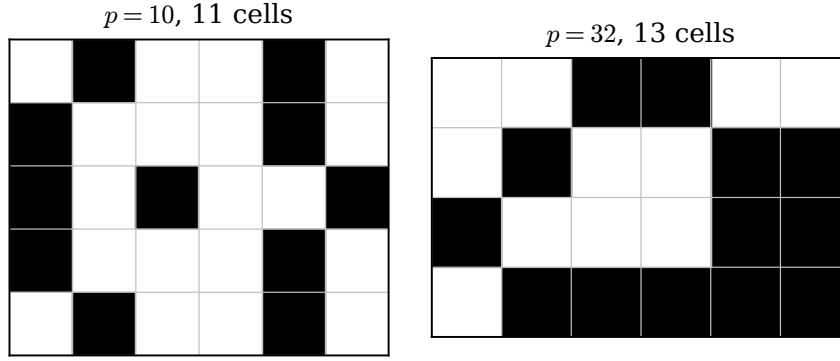


Figure 1: One phase of the period-10 and period-32 oscillators of B37/S2378.

Rule range. An oscillator works in every rule that agrees with B37/S2378 on the neighbour counts that actually occur during its cycle. Recording, over one period, which counts cause a birth or a survival and which counts occur without one gives the range. The period-10 oscillator works exactly in the rules from B37/S238 to B378/S2378, and the period-32 one from B37/S237 to B378/S2378. Both need birth on 7, so neither can exist in Conway’s rule or in any rule without B7.

Prior occurrence. As of 26 September 2026, Catagolue [4] has no census for any of the six rules in these two ranges, LifeWiki’s lists of Life-like rules [2, 3] do not list B37/S2378, and a code search of public GitHub repositories finds no pattern file for any of the six rule strings. We found no earlier description of these two oscillators; we could not search the ConwayLife forum, and a claim of novelty should be read with that in mind.

Machine-checked periods. The oscillators are encoded in Lean 4 [6] on a finite board wide enough that no live cell ever reaches its border during the cycle (margins of 4 and 7 cells), so the finite evolution coincides with the evolution in the plane. The following are proved by `decide`, i.e. by evaluation in the kernel, without `native_decide`:

- `p10_period`: 10 steps of B37/S2378 return the period-10 pattern to itself;
- `p10_period_minimal`: no smaller number of steps does;
- `p32_period`: 32 steps return the period-32 pattern to itself;
- `p10_not_conway`: under B3/S23 the period-10 pattern does not return to itself within 10 steps.

`#print axioms` reports only `propext`. The development builds with Lean v4.33.0 and a pinned Mathlib [7] and accompanies this note. The border condition is checked by our simulation, not by Lean, and the minimality of the period 32 is also computational (the census finds no earlier repetition).

3.3 Random soups

Random soups settle to a density of 0.122 under B37/S2378, against 0.052 under B3/S23 (Figure 2). The larger value is what one expects from survival on 7 and 8: dense clusters that B3/S23 would erode persist.

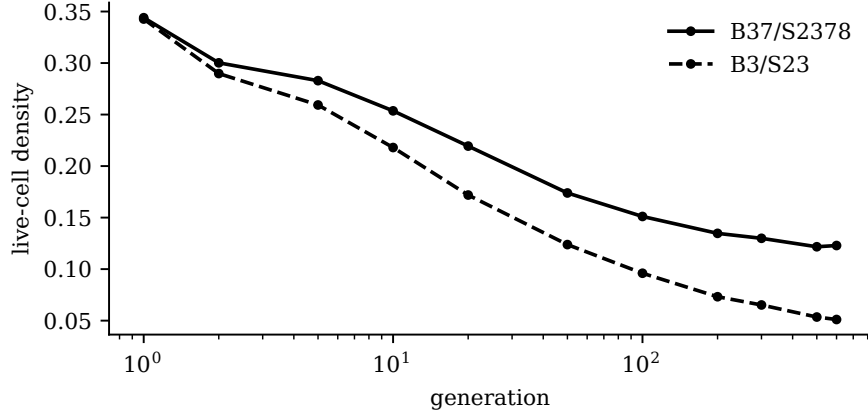


Figure 2: Live-cell density of random soups (mean of 40 soups of side 128, torus, initial density 0.30).

3.4 Unbounded growth

Of the 471 sampled unsettled seeds of B37/S2378, 338 are still growing after 1,000 generations, 71 are bounded but unsettled and 62 settle. Of the 465 sampled unsettled seeds of B3/S23, 6 are still growing after 1,500 generations; followed to 20,000 generations, all 6 settle, between generation 1,700 and 4,100. Under B37/S2378, 8 of the 10 seeds followed to 5,000 generations are still growing; fitting population $\propto t^\alpha$ between generations 1,000 and 5,000 gives α between 1.73 and 2.12 (median 1.85, $n = 8$).

The growth is a disk. For the three seeds of Figure 3 the core radius grows linearly, with speeds between 0.056 and 0.065 cells per generation, and the density inside the core stays between 0.111 and 0.130, the density that random soup settles to. The picture is therefore a region of settled-soup texture whose edge keeps converting empty space at a roughly constant speed, which makes the area, and so the population, quadratic in time; the fitted exponents below 2 reflect the early, not yet circular, stage. Gliders escape from the edge throughout.

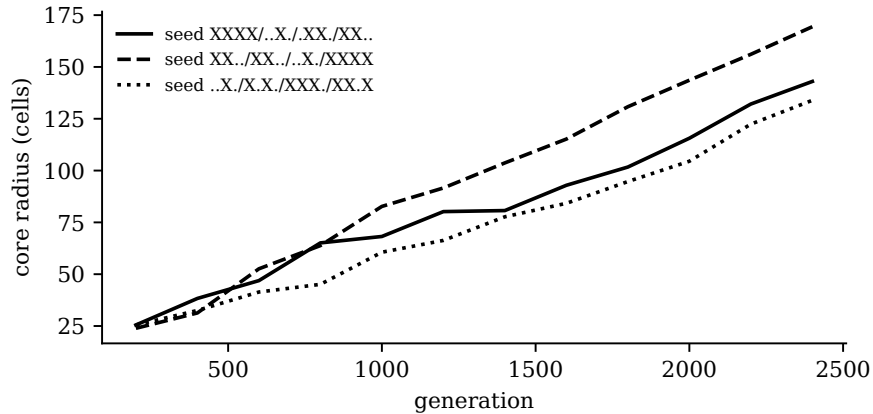


Figure 3: Core radius of three growing 4×4 seeds of B37/S2378 (blocks of 16×16 cells with density above 0.03).

3.5 Why there is no Catagolue census

Catagolue’s search program, *apgsearch* [4], runs each soup until it stabilises, with a budget of the order of 10^5 generations. Under B37/S2378 a typical soup does not stabilise: it grows as the disk above, so the work per generation grows with the square of the generation count. In our trial, a build of *apgsearch* configured for B37/S2378 had not completed 1,000 soups after nine minutes, where B3/S23 takes a fraction of a second. The rule is uncensused because the census method assumes soups that settle, not because it was overlooked.

4 Limitations

The census covers only the 4×4 box and only objects reached from it. The growth statistics come from samples (one seed in 25 and one in 20), and the front was measured on three seeds. The rule-range statement and the border condition for the Lean encoding are established by computation, not by proof. The novelty statement rests on the sources listed above.

Data and code

The census, soup, growth and front scripts, their JSON outputs, the script that generates every number, table and figure in this note from those outputs, the RLE files and the Lean development are deposited with this note and kept, with unit tests and continuous integration that rebuilds the Lean proofs and audits their axioms, at <https://github.com/joelcanary/life-like-rule-b37s2378>. Every number in the text is produced by that script; none is typed by hand.

Version history

v2 (26 September 2026). Corrects the number of seeds the census evolves: the 51,472 whose bounding box is exactly 4×4 , not all 65,535 non-empty patterns of the box as v1 stated, and the percentages derived from it. Adds the citations of Lean 4 and Mathlib and the link to the code repository. No other result changes.

Use of AI assistance

The computations, scripts and a first draft of this text were prepared with the help of an AI assistant (Claude, Anthropic). The author reviewed the results and is responsible for the content.

References

- [1] M. Gardner, *Mathematical Games: The fantastic combinations of John Conway’s new solitaire game “life”*, Scientific American 223 (1970), 120–123.
- [2] LifeWiki, *List of Life-like rules*, https://conwaylife.com/wiki/List_of_Life-like_rules (accessed 26 September 2026).
- [3] LifeWiki, *List of universal cellular automata*, https://conwaylife.com/wiki/List_of_universal_cellular_automata (accessed 26 September 2026).
- [4] A. P. Goucher, *Catagolue and apgsearch*, <https://catagolue.hatsya.com> (accessed 26 September 2026).

- [5] D. Eppstein, *Growth and decay in Life-like cellular automata*, in A. Adamatzky (ed.), *Game of Life Cellular Automata*, Springer (2010), 71–97; arXiv:0911.2890.
- [6] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in *Automated Deduction – CADE 28*, LNCS 12699, Springer (2021), 625–635.
- [7] The mathlib Community, *The Lean mathematical library*, in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, ACM (2020), 367–381.